STATISTICAL REPORT: 1,000-RUN MONTE CARLO BENCHMARK (*N* = 1,000)

Academic & Industry Evaluation of Seven Credit Card Fraud Detection Architectures

Authors: *Research Group | Python for Data Science (PDS) Capstone | IIIT Sri City*
Dataset: *ULB Machine Learning Group Benchmark (284,807 Transactions, 492 Frauds, 0.1727% Imbalance)*
Empirical Scope: *1,000 Stratified Splits | 7,000 Independent Model Fits | 56,962,000 Out-of-Sample Predictions*

Executive Summary & Core Empirical Findings

Evaluating fraud models on a single random train/test split creates severe estimator variance and partition luck. This standalone statistical monograph reports the definitive results of a **1,000-run Monte Carlo resampling benchmark (*N* = 1,000)** conducted across seeds $s \in \{1000, \ldots, 1999\}$. Every model architecture was re-trained from mathematical first principles on 1,000 distinct stratified partitions (7,000 total fits; 56,962,000 classifications) to audit resampling stability, tail-risk exposure (CVaR), and true operational loss trade-offs.

1. Overall Financial & Risk Champion: GBDT Gradient Boosted Trees achieves the lowest total net operational loss (**$4,369.26 ± $77.66** per split) under standard triage (C_{FP} = $2.50). It generates only **880.19 false positives** (1.55% FPR), achieving the benchmark's highest precision (**9.30%**), highest F1-score (**16.76%**), and lowest extreme tail risk ($CVaR_{99\%}$ = **$8,117.52**).	**2. Push Challenge Champion: Soft-Voting** The Multi-Paradigm Soft-Voting Ensemble maximizes discriminative power across all models (**ROC-AUC = 0.9808 ± 0.0005**), capturing **91.12% recall**. When alert costs are minimal ($C_{FP} \leq$ $0.64, e.g., in-app biometric push challenges), Soft-Voting minimizes net financial loss (**$2,485.42** at $0.50/alert).
3. The Recall Fallacy: Cost-Sensitive LogReg Cost-Sensitive LogReg maximizes event recall (**95.53%**) and dollar recall (**94.57%**), missing only $663.85 in fraud. However, it triggers an unbearable **8,444.07 false positives** per split (14.85% FPR). At $2.50/alert, triage costs explode to **$21,110.16**, driving net loss to **$21,774.01**—over $17,400 worse than GBDT.	**4. Analytical Breakeven Review Cost ($0.6358)** Equating the loss functions between Soft-Voting and GBDT reveals the exact breakeven triage cost threshold: C_{FP}^{*} = ($2,168.80 - $1,590.85) / (1,789.14 - 880.19) = **$0.6358**. For review costs below **63.6¢**, Soft-Voting is optimal; for costs above 63.6¢, GBDT strictly dominates.

1. Monte Carlo Simulation Protocol & Resampling Design

1.1 Resampling Architecture: Exactly N = 1,000 independent partitions were generated using pseudo-random seeds $s \in \{1000, \ldots, 1999\}$. Each partition enforces strict stratified 80/20 train/test splitting: **227,845 training transactions** (394 frauds, 227,451 legitimate) and **56,962 test transactions** (98 frauds, 56,864 legitimate). All seven models are fitted from scratch on the training set and evaluated strictly on the held-out test split, accumulating 56,962,000 out-of-sample inferences. Parallel multi-core execution on a 12-core CPU completed in 831.3 seconds (~13.86 min).

1.2 Central Limit Theorem & Confidence Intervals: Across N = 1,000 runs, the sample mean $\theta\blacksquare_m$ and sample standard deviation s_m yield a 95% CLT confidence interval half-width defined as:

$$\text{Margin}_{95\%} = z_{0.975} \cdot \frac{s_m}{\sqrt{N}} = 1.96 \cdot \frac{s_m}{\sqrt{1000}} \approx 0.0620 \cdot s_m$$

Critical Resampling Caveat (Dietterich, 1998; Bengio & Grandvalet, 2004): Because all 1,000 splits draw from the identical historical pool of 492 fraud cases, the partitions are not independent samples from an infinite superpopulation. The estimates are positively correlated due to shared instances. Consequently, reported confidence intervals represent *measures of empirical resampling stability and partition sensitivity across this dataset*, rather than asymptotic guarantees on future non-stationary transaction streams.

2. Plain-English Metric Interpretation Guide for Banking Operations

In extreme class rarity (0.172% fraud), standard symmetric machine learning statistics are deceptive. The following glossary translates statistical metrics into operational banking realities:

Metric & Formula	Statistical Meaning	Operational Banking Reality & Risk Interpretation
Event Recall (TPR) **TP / (TP + FN)**	Percentage of actual fraudulent transactions detected.	If 100 criminal attacks occur and recall is 90%, 90 are stopped and 10 slip through. Maximizing recall blindly causes severe investigator queue bloat.
Dollar Recall **$\sum_{TP}$ Amt / $\sum_{All}$ Amt**	Fraction of stolen monetary capital prevented from loss.	Card attacks range from $0.01 micro-pings to $4,000 wire purchases. Catching one $4,000 attack saves more money than catching 400 $5 charges.
Precision (PPV) **TP / (TP + FP)**	Trustworthiness of an alert: fraction of flagged cards that are truly fraud.	GBDT's precision of 9.30% means 1 out of 11 alerts is an actual crook. Cost-Sensitive LogReg's 1.11% precision means 90 out of 91 alerts are innocent cardholders.
False Discovery (FDR) **FP / (TP + FP) = 1 - Prec**	Proportion of false alarms inside the analyst investigation queue.	Cost-Sensitive LogReg forces analysts to waste 98.89% of their hours reviewing innocent accounts. GBDT drops queue waste to 90.70%, quadrupling efficiency.
False Positive Rate **FP / (FP + TN)**	Probability of flagging an innocent customer's transaction.	**Regulatory Distinction:** Model FPR (1.55% for GBDT) is distinct from PSD2 SCA TRA gross fraud ceilings (1 bps / 0.01%). FPR does not guarantee regulatory compliance.
Net Loss Function **(FP × $2.50) + $FN_{\$}$**	Total operational cost on the issuing bank's income statement.	Under US Reg E, the bank absorbs 100% of missed fraud ($FN_{\$}$). Each false alert costs $2.50 ($0.05 gateway + $1.75 labor @ $35/hr + $0.70 churn friction).
Tail Risk ($CVaR_{99\%}$) **$E[Loss \mid Loss \geq VaR_{99\%}]$**	Average institutional loss during the worst 1% of market scenarios.	Measures resilience against coordinated bot surges. GBDT caps worst-case loss at $8,117.52; Cost-Sensitive LogReg explodes to $27,284.65.

3. Master 1,000-Run Empirical Benchmark Summary (Table 3.1)

Values report the **Sample Mean ± 95% CLT Confidence Interval Half-Width** across all 1,000 independent simulation runs (56,962 test transactions per run):

Model Architecture	Event Recall	Dollar Recall	Precision	F1-Score	ROC-AUC	FPR (%)	Stolen $ Saved	False Pos (FP)	Missed Fraud	Net Loss ($)
1. Baseline LogReg	90.42% ± 0.18%	85.42% ± 0.56%	5.95% ± 0.04%	11.16% ± 0.08%	0.9780 ± 0.0006	2.50% ± 0.02%	$10,124 ± $127	1,420.6 ± 11.5	$1,771 ± $72	$5,322 ± $72
2. Cost-Sens. LogReg	**95.53% ± 0.13%**	**94.57% ± 0.40%**	1.11% ± 0.01%	2.18% ± 0.01%	0.9745 ± 0.0006	14.85% ± 0.09%	**$11,230 ± $136**	8,444.1 ± 50.1	**$664 ± $120**	$21,774 ± $120
3. Neural Net (MLP)	90.35% ± 0.18%	85.33% ± 0.57%	5.72% ± 0.05%	10.75% ± 0.08%	0.9702 ± 0.0007	2.61% ± 0.02%	$10,116 ± $128	1,485.8 ± 13.4	$1,778 ± $76	$5,493 ± $76
4. GBDT Classifier	88.43% ± 0.20%	82.02% ± 0.57%	**9.30% ± 0.11%**	**16.76% ± 0.18%**	0.9778 ± 0.0005	1.55% ± 0.02%	$9,725 ± $125	**880.2 ± 11.9**	$2,169 ± $78	**$4,369 ± $78**
5. Isolation Forest	60.46% ± 0.60%	57.48% ± 0.91%	8.22% ± 0.12%	14.40% ± 0.19%	0.9451 ± 0.0009	**1.21% ± 0.02%**	$6,827 ± $136	**688.8 ± 10.2**	$5,067 ± $122	$6,789 ± $122
6. Soft-Voting Ensem.	91.12% ± 0.17%	86.91% ± 0.54%	4.82% ± 0.03%	9.15% ± 0.06%	**0.9808 ± 0.0005**	3.15% ± 0.03%	$10,303 ± $128	1,789.1 ± 14.3	$1,591 ± $72	$6,064 ± $72
7. Stacked Meta-Learner	91.86% ± 0.17%	87.96% ± 0.53%	3.63% ± 0.02%	6.97% ± 0.04%	0.9793 ± 0.0005	4.26% ± 0.03%	$10,430 ± $129	2,421.5 ± 17.8	$1,464 ± $73	$7,518 ± $73

4. Master Arithmetic Reconciliation Audit (Unrounded Underlying Values)

To guarantee mathematical rigor, every model's reported net operational loss is audited against the exact financial cost equation down to the cent: **Total Net Loss = (FP■ × $2.50) + FN■$**.

Model Architecture	Mean False Pos (FP■)	Alert Cost (FP■ × $2.50)	Missed Fraud (FN■$)	Total Calculated Loss	Table 3.1 Benchmark	Reconciliation Status
1. Baseline LogReg	1,420.592	$3,551.48	$1,770.57	**$5,322.05**	$5,322 ± $72	**Exact Match**
2. Cost-Sens. LogReg	8,444.065	$21,110.16	$663.85	**$21,774.01**	$21,774 ± $120	**Exact Match**
3. Neural Net (MLP)	1,485.818	$3,714.55	$1,778.08	**$5,492.63**	$5,493 ± $76	**Exact Match**
4. GBDT Classifier	880.185	$2,200.46	$2,168.80	**$4,369.26**	$4,369 ± $78	**Exact Match (Min Loss)**
5. Isolation Forest	688.754	$1,721.88	$5,066.98	**$6,788.86**	$6,789 ± $122	**Exact Match**
6. Soft-Voting Ensem.	1,789.139	$4,472.85	$1,590.85	**$6,063.70**	$6,064 ± $72	**Exact Match**
7. Stacked Meta-Learner	2,421.549	$6,053.88	$1,464.31	**$7,518.19**	$7,518 ± $73	**Exact Match**

5. Tail-Risk & Empirical Loss Distribution Analysis

Average loss evaluates expected daily drag, but fails to capture vulnerability to criminal bot spikes or organized fraud waves. The table below details the full empirical loss distribution, skewness, kurtosis, and tail-risk metrics (VaR / CVaR) across 1,000 runs:

Model Architecture	Mean Loss ($)	Median Loss	StdDev ($)	$VaR_{95\%}$ ($)	$VaR_{99\%}$ ($)	$CVaR_{99\%}$ ($)	Skewness	Kurtosis
1. Baseline LogReg	$5,322.05	$5,117.42	$1,166.22	$7,474.71	$8,308.31	$8,934.98	+0.71	+0.18
2. Cost-Sens. LogReg	$21,774.01	$21,769.60	$1,937.71	$25,112.25	$26,449.77	$27,284.65	+0.16	-0.03
3. Neural Net (MLP)	$5,492.63	$5,262.89	$1,218.48	$7,950.18	$8,624.46	$8,942.64	+0.68	+0.01
4. GBDT Classifier	**$4,369.26**	**$4,194.31**	$1,252.96	**$6,768.28**	**$7,622.60**	**$8,117.52**	+0.62	+0.09
5. Isolation Forest	$6,788.86	$6,594.01	$1,973.57	$10,168.79	$11,741.74	$12,969.31	+0.50	+0.08
6. Soft-Voting Ensem.	$6,063.70	$5,858.76	$1,160.08	$8,281.75	$9,174.48	$9,396.16	+0.70	-0.03
7. Stacked Meta-Learner	$7,518.19	$7,315.78	$1,185.64	$9,773.20	$10,673.71	$10,885.98	+0.62	+0.01

6. Architectural Comparison & Mathematical Decision Surfaces

Architecture	Mathematical Decision Boundary	Optimization Objective	Imbalance Mitigation Strategy
1. Baseline LogReg	Linear Hyperplane: $z = w^T x + b$	Binary Cross-Entropy (BCE)	Natural empirical priors (symmetric threshold $\tau = 0.5$)
2. Cost-Sens. LogReg	Shifted Linear Hyperplane ($\Delta b \approx +6.36$)	Weighted Binary Cross-Entropy	Inverse class weighting ($w_1 \approx 578{:}1$ minority penalty)
3. Deep MLP (Adam)	Smooth Non-Linear Manifold (3 Layers, LeakyReLU)	Regularized BCE + Backprop	Adaptive moment scaling (m_t, v_t) with weight decay
4. GBDT Classifier	Axis-Aligned Hyper-Rectangles (Depth 3 Stumps)	Stage-wise Negative Gradient Fit	Shrinkage ($\eta = 0.1$) on greedy pseudo-residuals $y_i - p_{m\text{-}1}$
5. Isolation Forest	Random Enclosed Polyhedra (Unsupervised)	Average Isolation Depth $h(x)$	Completely label-free; isolates rare anomalies near tree roots
6. Soft-Voting Ensem.	Blended Posterior Surface: $1/3 \sum p_k(x)$	Minimizes Estimator Variance	Blends linear, deep, and tree manifolds to cancel uncorrelated errors
7. Stacked Meta-Model	Meta-Learner Hyperplane over OOF Probabilities	Meta-Level Log-Loss	5-Fold Out-of-Fold cross-validated meta-feature training

7. Multi-Tier Operational Cost Sensitivity & Breakeven Frontiers

Verification costs depend heavily on the customer interaction channel: an automated biometric push notification costs pennies, whereas a manual security telephone call costs multiple dollars. We evaluate total net financial loss across five distinct operational cost tiers:

Model Architecture	C_{FP} = \$0.50 (In-App Push)	C_{FP} = \$1.00 (SMS Alert)	C_{FP} = \$2.50 (Base Queue)	C_{FP} = \$5.00 (Manual Call)	C_{FP} = \$10.00 (Analyst Triage)	Optimal Operating Domain
1. Baseline LogReg	\$2,480.87	\$3,191.16	\$5,322.05	\$8,873.53	\$15,976.49	High explainability baseline
2. Cost-Sens. LogReg	\$4,885.88	\$9,107.91	\$21,774.01	\$42,884.17	\$85,104.49	Viable only if C_{FP} < \$0.16
3. Neural Net (MLP)	\$2,520.99	\$3,263.90	\$5,492.63	\$9,207.17	\$16,636.26	Automated digital queues
4. GBDT Classifier	\$2,608.89	**\$3,048.98**	**\$4,369.26**	**\$6,569.72**	**\$10,970.65**	**Optimal for C_{FP} ≥ \$0.64**
5. Isolation Forest	\$5,411.35	\$5,755.73	\$6,788.86	\$8,510.74	\$11,954.51	Zero-day anomaly sentinel
6. Soft-Voting Ensem.	**\$2,485.42**	\$3,379.99	\$6,063.70	\$10,536.55	\$19,482.24	**Optimal for C_{FP} ≤ \$0.64**
7. Stacked Meta-Learner	\$2,675.09	\$3,885.86	\$7,518.19	\$13,572.06	\$25,679.81	Complex calibrated cascades

Analytical Breakeven Derivations

By setting $\text{Loss}_A(C_{FP}^*) = \text{Loss}_B(C_{FP}^*)$, we solve for the exact review cost threshold where the optimal institutional model switches:

$$C_{FP}^* = \frac{\overline{FN}_{USD,A} - \overline{FN}_{USD,B}}{\overline{FP}_B - \overline{FP}_A}$$

1. Soft-Voting Ensemble vs. GBDT Classifier:
C_{FP}^* = (\$2,168.80 - \$1,590.85) / (1,789.14 - 880.19) = \$577.95 / 908.95 = **\$0.6358 (63.6¢)**.
Institutional Rule: When verifying an alert costs **less than 63.6 cents** (e.g., biometric push-to-confirm), Soft-Voting minimizes financial loss. When review costs **more than 63.6 cents** (analyst manual triage), GBDT strictly minimizes financial loss.

2. GBDT Classifier vs. Baseline Logistic Regression:
C_{FP}^* = (\$2,168.80 - \$1,770.57) / (1,420.59 - 880.19) = \$398.23 / 540.40 = **\$0.7369 (73.7¢)**.
Institutional Rule: GBDT outperforms Baseline LogReg whenever alert triage costs more than 73.7 cents.

3. Cost-Sensitive LogReg vs. Baseline Logistic Regression:
C_{FP}^* = (\$1,770.57 - \$663.85) / (8,444.07 - 1,420.59) = \$1,106.72 / 7,023.48 = **\$0.1576 (15.8¢)**.
Institutional Rule: Cost-Sensitive LogReg is economically unviable unless alert verification is virtually free (< 15.8 cents).

8. Architectural Deep Dive: Why Models Behave As They Do

Why GBDT Wins Overall: Trees partition feature space via orthogonal axis-aligned splits. Extreme outliers in transaction amounts cannot tilt the global boundary (unlike linear hyperplanes). GBDT isolates compact hyper-rectangles of high fraud density, resulting in conservative, well-calibrated posterior probabilities that suppress false alarms to 880.2—quadrupling investigator precision to 9.30%.
Why Soft-Voting Dominates ROC-AUC: Blending predictions across three completely distinct mathematical paradigms (linear hyperplane + smooth neural manifold + axis-aligned trees) averages out independent model variance: $p\blacksquare(x) = 1/3\ [p_{\text{LogReg}} + p_{\text{MLP}} + p_{\text{GBDT}}]$. The resulting ensemble achieves superior discriminative ranking (ROC-AUC = 0.9808), capturing 91.12% event recall.

9. Master Operational Decision Matrix: Which Model is Best for What?

No single machine learning model is globally superior across all banking environments. The matrix below maps each architecture to its optimal production operating domain based on empirical simulation results:

Model Architecture	Core Technical Strength	Primary Operational Vulnerability	Cost Regime	$CVaR_{99\%}$	Optimal Banking Deployment Scenario	Overall Verdict
GBDT Classifier	Highest precision (9.30%), lowest false alarms (880.2), tree robustness	Axis-aligned step boundaries require tree interpretability tooling	$C_{FP} \geq \$0.64$	**$8,117.52**	**Core Transaction Screening:** Production gating for card-not-present fraud & analyst queues.	**OVERALL CHAMPION** Lowest Net Loss
Soft-Voting Ensemble	Highest ROC-AUC (0.9808), smoothed variance, 91.12% recall	Higher alert volume than GBDT (1,789 alerts per split)	$C_{FP} \leq \$0.64$	$9,396.16	**Digital Push-to-Confirm:** In-app biometric challenges where customer friction is minimal.	**PUSH CHAMPION** Max Discriminative Power
Baseline LogReg	Full linear explainability, simple weight audit, sub-ms inference	Cannot capture non-linear cross-feature correlations	$C_{FP} \approx \$0.74$	$8,934.98	**Regulatory Gating:** High-velocity authorizations requiring automated Adverse Action codes.	**BEST FOR AUDIT** Zero Black-Box Scrutiny
Cost-Sens. LogReg	Maximum Event Recall (95.53%) and Dollar Recall (94.57%)	Severe alert explosion (8,444 alerts), extreme operational drag	$C_{FP} < \$0.16$	$27,284.65	**Catastrophic Wire Transfers:** Ultra-high-value payments (>$20k) where missed fraud is fatal.	**SPECIALIZED ONLY** Catastrophic Protection
Isolation Forest	Zero label supervision; detects novel structural anomalies	Lower supervised recall (60.46%), misses $5,067 in fraud	Zero-Day Sentinel	$12,969.31	**Unsupervised Watchdog:** Parallel fraud stream monitoring and cold-start rails.	**BEST SENTINEL** Zero-Day Anomaly Detection
Neural Net (MLP)	Models smooth non-linear manifolds across PCA components	Uncalibrated probabilities near 0 and 1; higher training cost	Digital Queues	$8,942.64	**Secondary Scoring Tier:** Deep multi-modal fraud screening alongside graph features.	**BALANCED** Non-Linear Baseline
Stacked Meta-Model	Blends meta-features via out-of-fold cross-validation	Alert inflation (2,421 FPs) without post-hoc threshold tuning	Tuned Cutoffs	$10,885.98	**Enterprise Stacks:** Production environments with post-hoc calibration layers.	**HIGH POTENTIAL** Meta-Learner

10. Two-Tier Production Screening Architecture (< 50ms Pipeline)

To meet the strict latency SLAs of payment networks (VisaNet, Mastercard Banknet < 50 ms), we recommend deploying a two-tier screening cascade combining linear velocity with non-linear precision:

[Incoming Payment Authorization Request]

- **Tier 1: Ultra-Fast Screening (< 1.0 ms)** — Evaluated via Vectorized Baseline Logistic Regression.
  - If $p(x) < 0.05 \rightarrow$ **AUTO-APPROVE** (97% of all transaction volume passed instantly with zero friction).
  - If $p(x) \geq 0.05 \rightarrow$ **ESCALATE TO TIER 2** (Remaining 3% volume routed to deep risk scoring).
- **Tier 2: Deep Risk Scoring (< 15.0 ms)** — Evaluated via GBDT Classifier + Soft-Voting Ensemble.
  - Dynamic Bayes Threshold: $p^*(x) = C_{FP} / [C_{FP} + \text{Amount}(x)]$.
  - If $p(x) < p^*(x) \rightarrow$ **APPROVE WITH AUDIT LOGGING**.
  - If $p(x) \geq p^*(x) \rightarrow$ **DYNAMIC INTERVENTION:** Low Amount (< $100) → 3DS Push Challenge; High Amount (> $1,000) → Decline & Analyst Triage.

11. Reproducibility & Research Artifact Directory

Artifact Verification: *Raw 1,000-run simulation dataset: benchmark_1000_runs_raw.csv (892 KB) | Summary data: benchmark_1000_runs_summary.csv | Statistical breakdown: benchmark_1000_runs_statistical_breakdown.csv | Execution script: run_1000_simulations.py | Worker module: simulation_worker.py | 20-Page Comprehensive Monograph: report/Credit_Card_Fraud_Detection_Comprehensive_Guide.pdf.*

References: *Bahnsen et al. (2015); Dal Pozzolo et al. (2014); Friedman (2001); Liu et al. (2008); Lin et al. (2017); Wolpert (1992); Davis & Goadrich (2006); Dietterich (1998); Bengio & Grandvalet (2004).*